

# MAJOR PROJECT REPORT

## Title: Online Payment Fraud Detection Using Machine Learning

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*Internship Program: Machine Learning Internship*

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*Domain: Machine Learning*

*Institution/Organization: Intrainz (Internspot)*

*Duration: Internship Training*

### 1. Introduction

With the rapid growth of digital payment systems such as credit cards, debit cards, UPI, and online banking, online financial transactions have become an essential part of everyday life. While these systems offer convenience and speed, they also increase the risk of fraudulent activities. Online payment fraud is a major concern for financial institutions, businesses, and users, as it can result in significant financial losses and reduced customer trust. Traditional fraud detection methods rely on rule-based systems, which are often inefficient in handling large volumes of transactions and adapting to new fraud patterns. Machine Learning (ML) provides an effective solution by learning patterns from historical transaction data and identifying suspicious activities automatically. This project focuses on detecting fraudulent online payment transactions using machine learning techniques.

## **2. Problem Statement**

The main objective of this project is to build a machine learning model that can accurately detect fraudulent online payment transactions. The system should be able to classify transactions as either legitimate or fraudulent based on transaction-related features. The challenge lies in the highly imbalanced nature of the dataset, where fraudulent transactions are very rare compared to legitimate ones.

## **3. Objectives of the Project**

The objectives of this project are:

- To understand the problem of online payment fraud detection
- To analyze transaction data using exploratory data analysis (EDA)
- To apply machine learning algorithms for fraud detection
- To evaluate the performance of the model using suitable metrics
- To build a reliable and efficient fraud detection system

## **4. Dataset Description**

The dataset used for this project is a credit card transaction dataset commonly used for fraud detection studies. The dataset consists of numerical features representing transaction details. For privacy and

security reasons, most features are transformed using Principal Component Analysis (PCA).

### **Dataset Features:**

- V1 to V28: Anonymized transaction features
- Time: Time elapsed between transactions
- Amount: Transaction amount
- Class: Target variable (0 = Legitimate, 1 = Fraud)

The dataset is highly imbalanced, with fraudulent transactions forming only a small fraction of the total data.

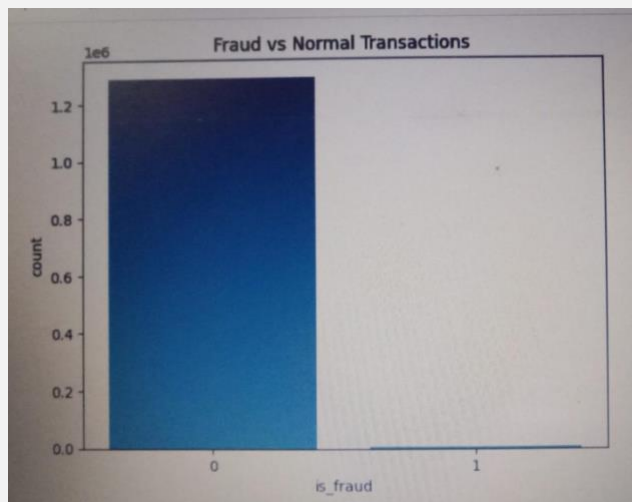
## **5. Methodology**

The methodology followed in this project includes the following steps:

- Data loading and preprocessing
- Exploratory Data Analysis (EDA)
- Feature and target selection
- Splitting the dataset into training and testing sets
- Model training using machine learning algorithms
- Model evaluation using performance metrics

## **6. Exploratory Data Analysis (EDA)**

EDA was performed to understand the distribution of data and identify important patterns. The class distribution revealed a large imbalance between legitimate and fraudulent transactions. Visualization techniques such as bar plots and heatmaps were used to analyze transaction behavior and model performance. EDA helps in identifying trends, anomalies, and relationships within the data, which improves the effectiveness of the machine learning model.



## 7. Machine Learning Model Used

### Logistic Regression

Logistic Regression is a supervised machine learning algorithm widely used for binary classification problems. In this project, it is used to classify transactions into fraudulent or legitimate categories.

Reasons for choosing Logistic Regression:

- Simple and efficient

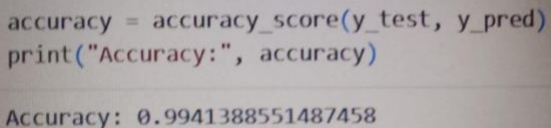
- Works well for binary classification
- Easy to interpret
- Suitable for large datasets

The model learns the relationship between input features and the target variable and predicts the probability of a transaction being fraudulent.

## 8. Model Training and Testing

The dataset was split into training and testing sets using an 80:20 ratio.

The Logistic Regression model was trained on the training data and evaluated on the testing data. The model learned patterns associated with fraudulent behavior from historical transaction data.



```
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)

Accuracy: 0.9941388551487458
```

## 9. Model Evaluation

The performance of the model was evaluated using the following metrics:

Accuracy: Measures the overall correctness of the model

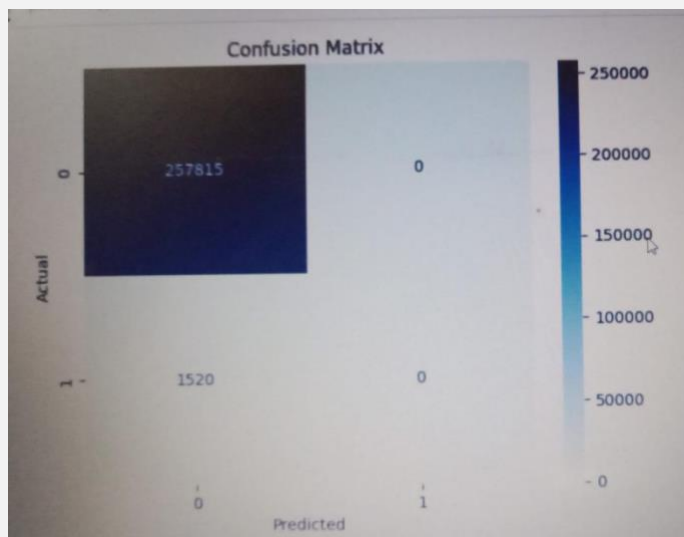
Precision: Indicates how many predicted fraud cases were actually fraud

Recall: Indicates how many actual fraud cases were correctly detected

F1-Score: Harmonic mean of precision and recall

Confusion Matrix: Shows true positives, true negatives, false positives, and false negatives

The evaluation results demonstrate that the model is capable of detecting fraudulent transactions effectively.



## 10. Results and Discussion

The Logistic Regression model achieved high accuracy on the test dataset. The confusion matrix showed that the model correctly classified most legitimate transactions while identifying fraudulent transactions with reasonable precision and recall. Although accuracy is high, other metrics such as precision and recall are more important due to the imbalanced nature of the dataset. The results confirm that machine learning can be effectively applied to detect online payment fraud and reduce financial losses.

## **11. Advantages of the System**

- Automated fraud detection
- Faster processing of transactions
- Reduced manual intervention
- Improved security for online payments
- Scalable and adaptable to new data

## **12. Limitations**

- Dataset is highly imbalanced
- Logistic Regression may not capture complex patterns
- Performance depends on quality of data
- Real-time deployment requires further optimization

## **13. Future Scope**

The project can be enhanced in the future by:

- Using advanced algorithms such as Random Forest, XGBoost, or Neural Networks
- Handling class imbalance using resampling techniques
- Implementing real-time fraud detection systems

- Integrating the model with banking or payment applications

## **14. Conclusion**

Online payment fraud detection is a critical problem in today's digital world. This project successfully demonstrated the use of machine learning techniques to identify fraudulent transactions. Logistic Regression proved to be an effective baseline model for fraud detection. With further improvements and advanced models, the system can be made more accurate and robust for real-world applications.

## **15. References**

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